CatFR5 Task Design Document

Approver	Signature	Date
Mike Kahana		

Version	Author	Description	Date
1.0	James Kragel	First version	07/29/2016

Overview

The CatFR5 task implements closed loop brain stimulation during a memory task to test the hypothesis that direct electrical stimulation (DES) of the brain during poor memory states results in better memory than a control condition with no brain stimulation. This work extends upon the prior experiment, CatFR3, by extending the application of brain state dependent stimulation to the retrieval phase of the task. This paradigm is dependent upon two additional tasks: CatFR1

and PS, in which biomarkers of successful retrieval are determined for individual patients (CatFR1), and optimal stimulation parameters are determined (PS).

In CatFR1 we recorded brain activity from multiple implanted electrodes during the encoding and retrieval of word lists and identified neural correlates of good and poor retrieval states (biomarkers).

We sought to predict whether memory search would lead to successful (correct recall of an item from the prior list) or unsuccessful (no recall of an item in the subsequent 2 sec) based on the spectral decomposition of the time-series of bipolar neural recordings from all N channels recorded. We thus fit a L2-regularized logistic regression model to the data in which the LHS variable was the successful/unsuccessful status of a period of memory search and the RHS variables were the wavelet power in each of 8 frequency bands and across all N channels (henceforth classifier). The model was fit to data from sessions 1 through M-1 and validated on session M, with the penalty parameter determined based on the distribution of best fitting parameters across all patients who performed the experiment excluding the patient whose data were being fit. The model was able to reliably predict recall based on the spectral decomposition of the recall period data, as described by Kragel et al. (submitted).

The present study (CatFR5) aims to directly test the hypothesis that direct electrical stimulation (DES) delivered during poor retrieval states can produce reliably better memory than observed on control lists without any DES. On half of the experiment lists, DES will be under control of an algorithm that evaluates the state of the brain during epochs of memory search and decides whether or not to stimulate vs. control lists in which no stimulation is provided.

Methods

Except as noted, the design of the behavioral task and the procedure for it are identical to FR1. The reader is referred to that design document for further details. Subjects will perform delayed free recall with categorized stimuli on 26 study-test lists, the first of these being practice (all data from all lists, including the practice list, will be recorded). Stimulation is not delivered to the brain during lists 1-4 as these lists are used to evaluate the ability of the logistic-regression classifier, trained on the same patient's FR1 data, to predict the recalled status of items in FR5. Following the first 4 lists, the remaining 22 lists will be divided into two conditions: Stim (16 lists) and NoStim lists (6 lists). These 22 lists will be pseudorandomly interleaved to balance the number of Stim and NoStim lists in the first and second halves of the session.

ECoG/iEEG Analysis

During stimulation lists, EEG data are analyzed to determine when to turn on the stimulator. The raw EEG data is converted from the default referential recording to a bipolar reference between neighboring electrodes as determined by the bipolar montage stored in each patient's '*Talairach structure*'[1]. These bipolar recordings are then filtered to remove 60Hz line noise using

[1] *Talairach structure* refers to the eeg_toolbox/PTSA data object used by the RAM team to represent information about each patient's montage. For the purposes of FR5, the relevant information is the set of bipolar electrode pairs,

a Butterworth stop band filter (58-62Hz range, fourth order). A Morlet wavelet transformation is then applied to the signal (wave number = 5) to estimate spectral power during word encoding periods (8 wavelets log-spaced between 20-200Hz; center frequencies: [20.0, 27.9, 38.6, 53.7, 74.6, 103.6, 143.94, 200.0] Hz). For analysis of both the training (FR1) and closed-loop data (FR5) we use power from 600-100 ms prior to recall onset with a truncated mirrored buffer that avoids repeating the first and last samples of the central interval in the buffers (262 ms duration buffers; total required data window = 1024 ms). Given our lowest frequency wavelet (20 Hz), the minimum required buffer size to avoid convolution edge artifacts is given by one half of the length of the wavelet = 139.26 ms[2]; additional buffer duration is included to maximize computational efficiency of computing power.

After applying the wavelet transformation we log-transform raw power values prior to removal of the buffer. These log transformed power values are then z-transformed to normalize them based on the within-session mean and standard deviation across all search period events (i.e., correct recalls and baseline events) in the session. The resulting z-transformed log power values are then features for the pattern classifier described below.

L2-logistic regression classifier: Prior to the start of FR5 we will have fit a regression model to data from multiple FR1 sessions as described below. This regression model will determine the timing of stimulation in FR5 stim lists. Using the spectral features described above we will estimate an L2-penalized logistic regression model to predict whether a search period leads to a correct recall or not (a proxy for good memory). Before estimating the beta weights (coefficients) on the input features one must specify a penalty parameter to be used in the estimation routine. We determine the penalty parameter based on the penalty value that leads to the highest average AUC in all prior RAM subjects with at least two FR1 sessions. The penalty parameter will be inversely weighted according to the class imbalance (Hastie et al., 2001). Class weights for each class are computed as $(1/n_A)/((1/n_A + 1/n_B)/2)$ where n_A is the number of events class A events, and n_B is the number of class B events. This ensures best fit values of the cost parameter, C, are comparable across different class distributions. For example, if the proportion of observations in classes A/B are .25 and .75 respectively, then the ratio of the weights for the penalty parameters for each class will be 3:1. The weight for class A will be .25/((.25+.75)/2) = .5, and the weight for class B will be .75/(.25+.75)/2 = 1.5. As new participants are collected and analyzed, their best penalty parameter will be added to the distribution of all previous patients.

Criterion for progressing from FR1 to FR5. Each patient's classifier will be tested for out-of-sample performance using leave-one-session-out (LOSO) cross validation. The true LOSO AUC will be compared to an AUC distribution (200 samples) generated from classification of label-permuted input data. $P < 0.05$ one-tailed will be used as the significance threshold for continuing to CatFR5. At least three sessions with a minimum of 15 lists per session are

not the coordinate system used to represent the electrode spatial locations. The structure in fact contains electrode information in Montreal Neurlogical Institute (MNI) space; the 'Talairach' nomenclature has nonetheless been maintained from earlier versions of the toolboxes.

[2] One half of the length of the wavelet is given by $n_z s f_s$ where n_z = number of standard units on the Gaussian; s = standard deviation of the Gaussian wavelet window; and f_s = sampling frequency.

required. If classifier is not significant with three sessions, additional CatFR1 sessions should be collected.

Closed-loop Procedure

During the retrieval phase of the StimLists in CatFR5 we will evaluate the classifier output during the 500 ms interval in a valid decoding period to decide whether to apply stimulation. If the classifier output during this interval is below a threshold value we will stimulate immediately. We will use Youden's *J* statistic to determine the threshold for stimulation individually for each patient. This statistic is given by *J* = Sensitivity + Specificity − 1 and can be represented visually as the height of the ROC curve above the unity (chance) line. We select the maximal *J* value across all points in the patient's CatFR1 ROC curve as our threshold for stimulation (Schisterman et al., 2005). This is equivalent to the threshold that maximizes classifier performance for detecting true positives while minimizing false alarms.

Data Normalization Process. The values that are input to the real-time classifier will be normalized to ensure the range of the data match the range expected by the classifier model. This will be done using the mean and standard deviation across word encoding events in the first four word lists (one practice list and three NoStim lists). Mean and standard deviation estimates will be updated as more data are collected. Updates will occur after every NoStim list (following the recall period). To do so, StimControl.m will need state information about timing of vocalization (including correct recalls, recall errors, and additional vocalization) onset and offsets.

Classifier Validation. To validate the classifier for use in CatFR5, the output from the classifier for each item presented on the first four lists will be compared to the classifier output when tested on data from all previous sessions using a two-sample Kolmogorov-Smirnov test. The null hypothesis that the CatFR5 and training data come from the same distribution must not be rejected (i.e., $P > 0.05$) for the CatFR5 task to continue. If this hypothesis is rejected, an error will be thrown and the session will proceed as an CatFR1 session. StimControl.m will also output (to the command line and to a log file on disk) the results of the two-sample Kolmogorov-Smirnov test comparing the distributions.

Stimulation Artifacts. We will avoid decoding brain data during stimulation. We will accomplish this by having two constraints for decoding. First, decoding will be performed following at least 1024 ms (i.e., one full data window) after the offset of either a vocalization response, or a previous stimulation event (see *Stimulation Onset Timing* below). Using this approach, when a stimulation decision is made, 500 ms of stimulation will be immediately delivered.

Stimulation Onset Timing. A stimulation waveform, specified by the Parameter Search (PS) task (see below), will be triggered when the multivariate classifier detects that the classifier output is below threshold, following a specified decoding interval. Specifically, stimulation decisions will be made based on classifier decoding of epochs within the memory search phase of the task. If

the classifier output for the decoded epoch falls below threshold, a stimulation waveform will be triggered immediately.

Selection of Closed Loop Stimulation Parameters

We will use the *J* statistic from all of a given patient's CatFR1 sessions to determine threshold (*K*) for identifying poor memory states in all PS2 and PS3 task sessions (e.g., bottom 30$_{th}$ percentile, etc.). Across all such-identified poor memory states we will find the parameter set, $\vec{\theta}$ across all PS sessions that maximizes the t-statistic comparison between post-pre classifier output and the FR1 control comparison set. Based on the parameter manipulated across PS2 and PS3 tasks we expect to identify a set theta that includes:

- Stimulation location (bipolar electrode pair)
- Amplitude **(SafeAmp-.5, SafeAmp-.25 or SafeAmp mA)**
- Pulse frequency **(10, 25, 50, 100, 200 Hz or single pulse)**
- Theta pattern frequency **(3, 4, 5, 6, 7, 8 Hz or n/a)**
- Theta train frequency **(100, 200 Hz or n/a)**

In the absence of patient-specific PS data, $\vec{\theta}$ will be chosen according to the best group-average parameters from the vault of PS datasets in existence at the time of the patient's participation. K will be determined based on the patient's own CatFR1 data across all sessions using the J-statistic as above.

Other Stimulation Parameters:

- Based on the aggregate PS1 data pulse duration will be set to 500ms
- Bipolar stimulation
- Pulse width: 600 microsecond pulse (300 us per phase)

References

- Hastie. T., Tibishirani, R., & Friedman, J. (2001). The elements of statistical learning: data mining, inference and prediction. *New York: Springer*.
- Schisterman, E.F.; Perkins, N.J.; Liu, A.; Bondell, H. (2005). "Optimal cut-point and its corresponding Youden Index to discriminate individuals using pooled blood samples". *Epidemiology* **16**: 73–81.

Here are some patients had open rather than closed loop stimulation delivered, including a column indicating whether they were open loop (randomstim=1).

subject	task_spec	stimchans	randomstim
R1124J	FR3	LC1-LC2	0
R1145J	FR3	RK2-RK3	0
R1154D	FR3	LTCG29-LTCG30	0
R1161E	FR3	L13HD1-L13HD2	0
R1163T	FR3	LC8-LC9	0
R1166D	FR3	LFPG3-LFPG4	0
R1170J	FR3	L7ST7-L7ST8	0
R1195E	FR3	3LD6-3LD7	0
R1195E	FR3	3LD8-3LD9	0
R1200T	FR3	LB7-LB8	0
R1201P	FR3	AT3-AT4	0
R1202M	FR3	RSO3-RSO4	0
R1202M	FR3	RSF2-RSF3	0
R1204T	FR3	LI5-LI6	0
R1217T	FR3	LC9-LC10	0
R1223E	FR3	7LD7-7LD9	0
R1230J	FR3	LA10-LA11	0
R1235E	FR3	1LD7-1LD9	0
R1236J	catFR3	LS4-LS5	0
R1293P	FR3	LA7-LA8	0
R1226D	catFR3	LASTG2-LASTG3	0
R1243T	catFR3	LB8-LB9	0
R1247P	catFR3	LC2-LC3	0
R1260D	catFR3	LLTG22-LLTG23	0
R1264P	catFR3	LA7-LA8	0
R1264P	catFR3	LB7-LB8	0
R1274T	catFR3	LC7-LC8	0
R1277J	catFR3	RMFC3-RMFC4	0
R1286J	catFR3	LA9-LA10	0
R1175N	PAL3	LPT5-LPT6	0
R1196N	PAL3	ALT5-ALT6	0
R1232N	PAL3	LST3-LST4	0
R1263C	PAL3	HCB10-HCB11	0
R1263C	PAL3	HCH7-HCH8	0
R1263C	PAL3	HCT10-HCT11	0
R1278E	PAL3	9LD2-9LD3	0
R1278E	PAL3	11LD3-11LD4	0
R1278E	PAL3	7LD3-7LD4	0
R1283T	PAL3	LI8-LI9	0
R1284N	PAL3	G22-G23	0
R1297T	PAL3	LC7-LC8	0
R1275D	FR5	RHD9-RHD10	1
R1292E	FR5	1LD9-1LD10	1
R1304N	FR5	PD5-PD6	1
R1308T	FR5	LB6-LB7	1
R1308T	FR5	LC7-LC8	1
R1308T	FR5	LB5-LB6	1
R1315T	FR5	LC6-LC7	1
R1317D	FR5	LTG19-LTG20	1
R1320D	FR5	RAD8-RAD9	1
R1321M	FR5	RF1-RF2	1
R1321M	FR5	RN1-RN2	1
R1323T	FR5	LC7-LC8	1
R1330D	FR5	LAD9-LAD10	1
R1339D	FR5	LHCD7-LHCD8	1
R1339D	FR5	LHCD6-LHCD7	1
R1341T	FR5	LT9-LT10	1
R1345D	FR5	LTG27-LTG28	1
R1345D	FR5	LTG12-LTG20	1
R1351M	FR5	RTG21-RTG22	1
R1380D	FR5	LAD7-LAD8	0
R1384J	FR5	RPMM11-RPMM12	0
R1384J	FR5	LPMI10-LPMI11	0
R1330D	catFR5	LAD7-LAD8	1
R1334T	catFR5	LC5-LC6	1
R1374T	catFR5	LA7-LA8	0

R1374T catFR5 LC6-LC7 0
R1375C catFR5 HCB6-HCB7 0
R1378T catFR5 LC8-LC9 0
R1379E catFR5 3Ld8-3Ld9 0
R1380D catFR5 LHCD8-LHCD9 0
R1383J catFR5 ROTWO11-ROTWO12 0
R1383J catFR5 RF8-RF9 0
R1383J catFR5 LB7-LB8 0
R1384J catFR5 LF7-LF8 0
R1384J catFR5 RFI9-RFI10 0
R1385E catFR5 3Ld8-3Ld9 0
R1387E catFR5 7Ld6-7Ld7 0
R1387E catFR5 9Ld6-9Ld7 0
R1387E catFR5 3Ld8-3Ld9 0
R1389J catFR5 STG6-STG7 0
R1390M catFR5 LTG15-LTG16 0
R1391T catFR5 LB7-LB8 0
R1395M catFR5 RSA4-RSA5 0
R1395M catFR5 RSA10-RSA11 0
R1395M catFR5 RC12-RC13 0
R1395M catFR5 RC11-RC12 0
R1398J catFR5 LB11-LB12 0
R1401J catFR5 STG3-STG4 0
R1406M catFR5 LTG26-LTG27 0
R1441T catFR5 LX11-LX12 0
R1465D catFR5 LHCD15-LHCD16 0
R1477J catFR5 LP7-LP8 0
R1487T catFR5 LC8-LC9 0
R1487T catFR5 LA7-LA8 0
R1488T catFR5 LA7-LA8 0
R1489E catFR5 1Ld13-1Ld14 0
R1489E catFR5 1Ld12-1Ld13 0
R1491T catFR5 LC7-LC8 0
R1498D catFR5 LHCD10-LHCD11 0
R1312N PAL5 G29-G30 0
R1318N PAL5 LAIIH2-LAIIH3 0
R1333N PAL5 LPLT2-LPLT3 0
R1365N PAL5 LPL4-LPL5 0
R1397D FR6 LACD8-LACD9 0
R1409D FR6 LAD11-LAD12 0
R1409D FR6 LHCD12-LHCD13 0
R1409D FR6 LHCD13-LHCD14 0
R1409D FR6 LAD12-LAD13 0
R1463E FR6 13Ld6-13Ld7 0
R1466J FR6 LFC7-LFC8 0
R1400N catFR6 TG13-TG14 0
R1400N catFR6 TG23-TG24 0
R1400N catFR6 TG15-TG16 0
R1403N catFR6 LF1-LF2 0
R1404E catFR6 19Ld8-19Ld9 0
R1409D catFR6 RACD7-RACD8 0
R1456D catFR6 LPHD9-LPHD10 0
R1461T catFR6 LI8-LI10 0
R1461T catFR6 LB7-LB9 0
R1461T catFR6 RC8-RC10 0
R1463E catFR6 3Ld6-3Ld7 0
R1474T catFR6 LC8-LC9 0
R1493T catFR6 LF10-LF11 0
R1493T catFR6 LF13-LF14 0
R1495J catFR6 ROc12-ROc13 0
R1495J catFR6 ROc11-ROc12 0
R1497T catFR6 LB8-LB9 0